Sean Lumasag

862-241-8905 | seanlumasag15@gmail.com | linkedin.com/in/seanlumasag | github.com/seanlumasag | seanlumasag.github.io

Education

Rutgers University — New Brunswick, NJ
B.S. in Computer Science — *Expected December 2027*

- GPA: 3.90 | Coursework: Data Structures & Algorithms, Database Systems, Computer Architecture

Experience

Software Engineer Intern — May 2026 – Present
Heller Industries — *Florham Park, NJ*

- Implemented a C#/.NET process-monitoring engine by applying eight Nelson rules to telemetry windows, identifying abnormal heating and conveyor patterns for dashboard investigation
- Built an oven-configuration importer by validating JSON setups and translating equipment geometry into shared WPF models, supporting standard and vacuum layouts through shared equipment and profile models
- Developed a temperature-profile comparison tool by interpolating measurements between misaligned sensor positions, generating five process metrics and aligned temperature-difference comparisons

Machine Learning Research Assistant — January 2026 – May 2026
WINLAB, Rutgers University — *Piscataway, NJ*

- Developed Python/PyTorch evaluation scripts by varying model-routing configurations across local and remote inference, collecting classification accuracy and edge-offloading results for mentor review
- Compared trained and random model-routing policies by evaluating 10,000 CIFAR-10 images at similar edge-use rates, measuring 92.42% versus 89.10% classification accuracy while offloading about three-quarters of requests

Software Engineer Intern — May 2025 – August 2025
Data Storytelling — *Milltown, NJ*

- Improved a PayPal ticket-purchase workflow by persisting captured payments before propagating transaction IDs into ticket records, preventing ticket creation on failed inserts and linking successful purchases to receipts
- Implemented a raffle-winner lookup by reloading Supabase winner records into React state when buyers revisit the site, restoring previous drawing results across sessions so users can confirm whether their tickets won
- Developed purchase-confirmation downloads by generating PDF receipts and calendar exports from payment, ticket, and drawing details, providing buyers with portable purchase records and reminders for scheduled raffle drawings

Projects

URL Shortener | *Java, Spring Boot, PostgreSQL, Redis*

- Developed a Spring Boot URL-shortening service by securing link-management APIs with JWT authentication, allowing registered users to create custom short links and inspect redirect activity through click analytics
- Implemented cached URL redirects by checking Redis for destinations before falling back to PostgreSQL, serving cached links without skipping the durable click events needed for user-facing redirect analytics

Image Processor | *TypeScript, NestJS, BullMQ, Redis, PostgreSQL*

- Built a batch image processor by dispatching transformations to Redis-backed BullMQ workers, supporting per-image resizing and WebP conversion without making API requests wait for background processing to finish
- Configured image-job retry handling by applying exponential backoff to failed BullMQ tasks, allowing repeat conversion attempts while PostgreSQL-backed job records expose the latest processing state to users

Live Chat | *Java, Spring Boot, React, PostgreSQL, WebSockets*

- Created a real-time chat application by publishing WebSocket messages to subscribed conversation rooms, supporting direct, group, and public messaging with live updates and stored conversation histories
- Enforced persist-before-broadcast messaging by committing chat messages through JPA transactions before publishing live WebSocket updates, retaining conversation history and state after application restarts

Technical Skills

Languages: Java, C#, Python, TypeScript, SQL
Frameworks: Spring Boot, .NET Framework, NestJS, React
Tools: PostgreSQL, Redis, BullMQ, Git, Docker